

Symmetric GaN phase-leg building block

$2 \times \text{EPC2361}$ per switch, gate driver in the middle, $75 \text{ V} / 100 \text{ A}_{\text{rms}}$

PCB-Design project, KTH

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Abstract

A symmetric phase-leg building block is proposed for a 75 V , $100 \text{ A}_{\text{rms}}$ inverter: two EPC2361 eGaN FETs in parallel per switch, one LT8418 half-bridge driver placed on the symmetry line, and an inner-layer (vertical) power loop for each FET pair. The DC link, auxiliary circuits and components are sized analytically; the copper parasitics of the proposed layout are extracted with Ansys Q3D; an LTspice double-pulse model built on the extracted matrix is used to select the gate resistors ($R_{\text{g,on}} = 2 \Omega$, $R_{\text{g,off}} = 0 \Omega$) and to compute the switching energy. At the rated point (2.65 kW per leg, 50 kHz) the estimated loss is 12.49 W , i.e. an efficiency of 99.53% against the 99.5% target (13.3 W budget); with maximum $R_{\text{DS(on)}}$ it is 99.46% . The extraction also showed that the PCB copper is the second-largest loss: 2 oz inner layers are needed (copper loss $4.67 \text{ W} \rightarrow 3.30 \text{ W}$).

1 Requirements and assumptions

DC link	75 V
Phase current	$100 \text{ A}_{\text{rms}}$, 141 A peak (sinusoidal)
Devices	EPC2361 (100 V , $0.75/1.0 \text{ m}\Omega$ typ/max at 25°C), 2 in parallel per switch
Efficiency	$\geq 99.5 \%$ per phase leg
Use	one phase leg of a three-phase two-level inverter (three blocks on one DC bus)
Rated point (efficiency)	$M = 1.0$, $\cos \varphi = 1 \Rightarrow P = 2.65 \text{ kW}$ per leg, loss budget 13.3 W
Switching frequency	50 kHz nominal, 100 kHz evaluated
Junction temperature for $R_{\text{DS(on)}}$	100°C (datasheet Fig. 9: $\times 1.48$)
DC-link ripple target	$\pm 2 \%$ (3 V peak-peak) at 50 kHz , worst case over M and $\cos \varphi$
PCB	4 layers, 1.6 mm , 2 oz on all layers, dielectrics $0.1 / 1.2 / 0.1 \text{ mm}$

The three-phase assumption matters for the DC link: with a common carrier the switching-frequency currents of the three legs partly cancel on a shared bus, and the fundamental-frequency power is constant, so no low-frequency bulk capacitance is needed. A single leg used on its own (half-bridge with a split DC link) needs far more capacitance (Section 3).

A reference point exists: EPC's EPC9186HC2 motor-drive board uses the same two EPC2361 per switch for $130 \text{ A}_{\text{rms}}$ up to 100 kHz , with $24 \times 10 \mu\text{F}/100 \text{ V}$ 1210 MLCCs per phase as its DC link.

2 Building-block concept

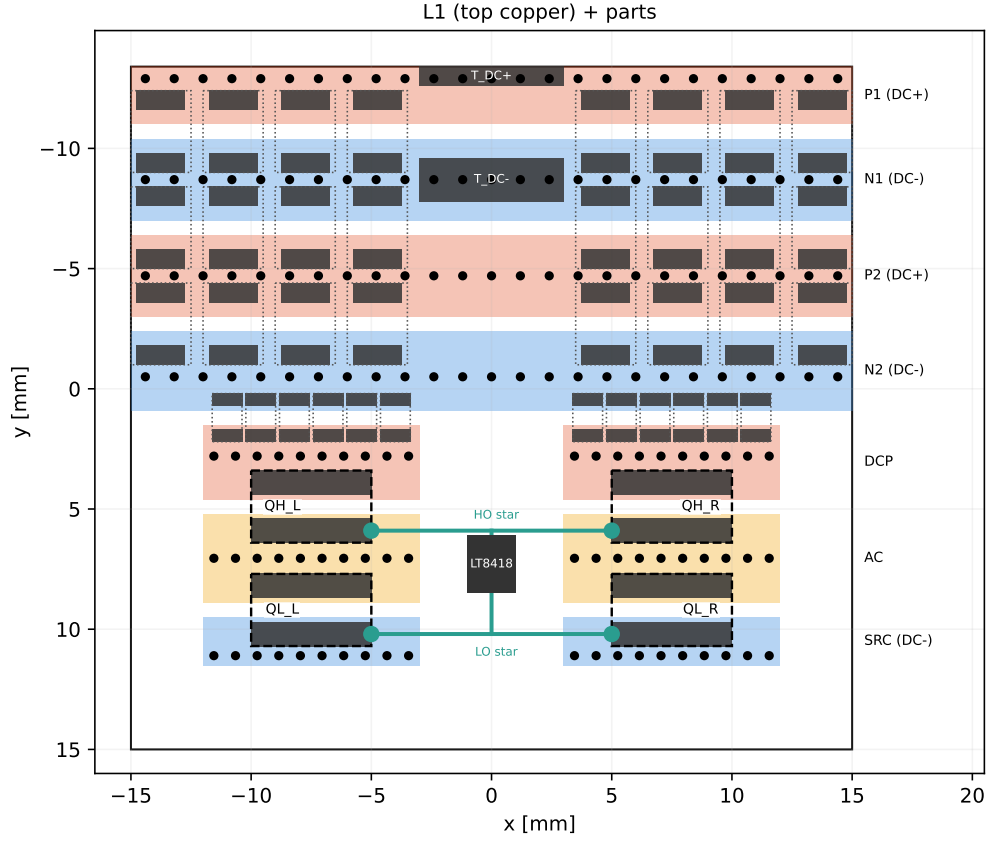


Figure 1: Top layer of the building block (30×28 mm). Two mirrored vertical power cells (QH over QL, local MLCCs next to QH) sit at $x = \pm 7.5$ mm; the LT8418 driver is on the symmetry line in a 6 mm corridor; HO and LO each drive both FETs through an equal star. A 24-part MLCC bank sits above the cells between alternating DC+/DC- strips. Dark rectangles are the pads used as Q3D terminals; dashed outlines are component bodies.

Every path that matters for current sharing is mirrored about $x = 0$: the two commutation loops, the HO and LO gate stars, the DC+ and DC- terminal feeds (on the centre line) and the AC terminal (bottom centre). The previous studies showed why: a 2.0/7.7 nH gate-loop mismatch gave a 19 % turn-on current imbalance and a 4:1 split of turn-on energy, and unequal common-source inductance was worse still. Kelvin gate returns are taken at each FET's source pad, so the source inductance is in the power loop only.

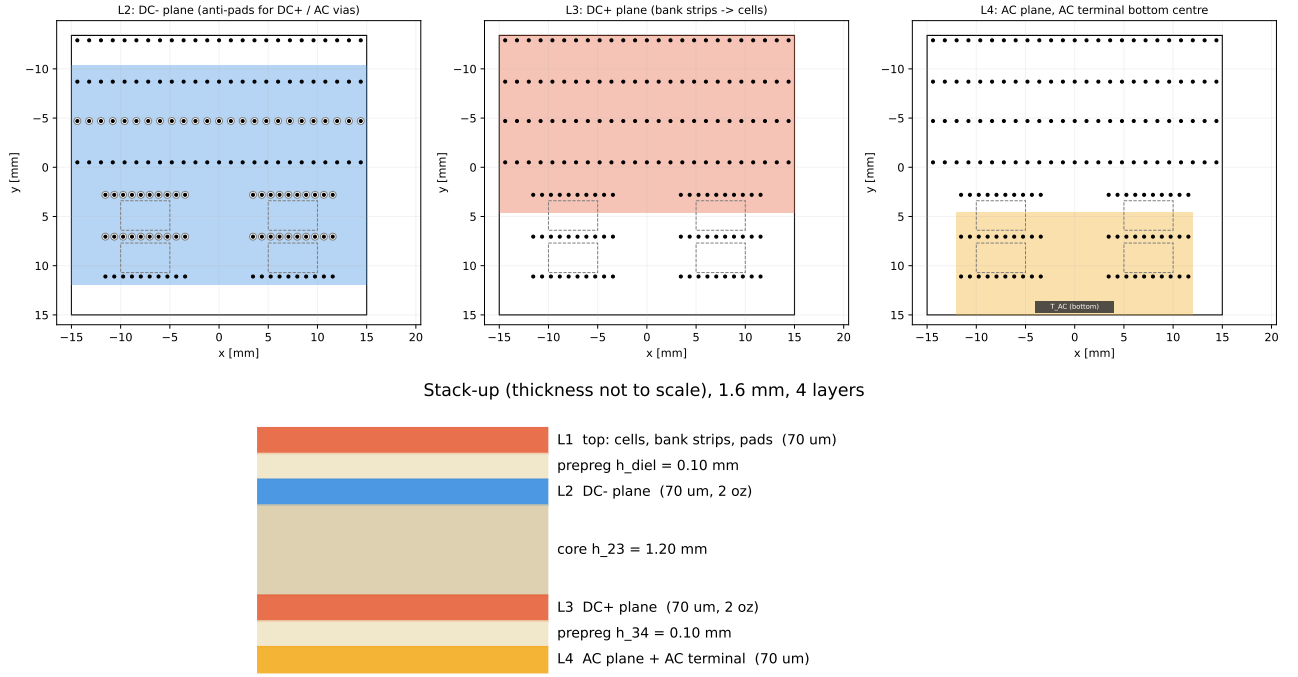


Figure 2: Inner and bottom layers and the 4-layer stack-up (2 oz on all layers, see Section 4). L2 is a solid DC–return 0.1 mm under the cells (anti-pads only where DC+ and AC vias pass); L3 carries DC+ from the bank strips to both cells; L4 joins both switch nodes to the AC terminal on the bottom side.

3 DC link and auxiliary circuits

The DC-link current was computed with an ideal three-phase SVPWM inverter (common triangular carrier, 50 kHz, 100 A_{rms}) over $M = 0.1\text{--}1.15$ and $\cos \varphi = 0\text{--}1$ (scripts/bb_sizing.py).

- HF ripple current of one leg alone: up to 50 A_{rms} (low M , any $\cos \varphi$); 29 A_{rms} at the rated point. This is what a block’s own capacitors must be rated for if blocks are physically separated.
- Three legs sharing one bus: up to 65 A_{rms} in total ($M \approx 0.6$, $\cos \varphi = 1$).
- Effective capacitance for $\pm 2\%$ ripple at 50 kHz: 233 μF for the shared bus, i.e. 78 μF per block; a single leg on its own would need 236 μF . At 100 kHz these halve.

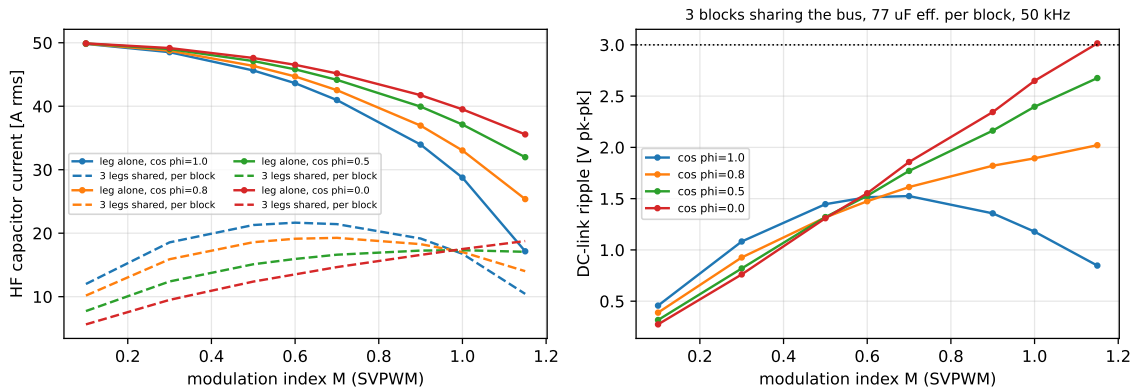


Figure 3: Left: HF capacitor current per leg (solid: leg alone; dashed: shared bus, per block). Right: DC-link ripple with the selected bank (77 μF effective per block) on a shared bus.

Selected per block: $24 \times$ Murata GRM32EC72A106KE05L (10 μF , 100 V, X7S, 1210) and $12 \times$ TDK C2012X7S2A105K125AB (1 μF , 100 V, X7S, 0805, six per cell next to QH). Assuming 30 % and 45 % of nominal capacitance at 75 V (to

be checked with the manufacturers' DC-bias data) this gives $77\ \mu\text{F}$ effective, $3.0\ \text{V}$ peak-peak on a shared bus at $50\ \text{kHz}$, and $9.1\ \text{V}$ for a leg alone. A stand-alone leg therefore needs additional bulk capacitance (film or polymer) on the DC bus. The bank ESR loss is $0.39\ \text{W}$ at the rated point.

Function	Part	Sizing
Power switches	$4 \times \text{EPC2361}$	2 per switch, I_{rms} per device $35.4\ \text{A}$
Gate driver	ADI LT8418 ($100\ \text{V}$, $4\ \text{A}/8\ \text{A}$, split outputs, $1.5\ \text{ns}$ matching)	one per block, on the symmetry line
Gate resistors	0402 thin film, per FET $R_{\text{g,on}} = 2\ \Omega$, $R_{\text{g,off}} = 0\ \Omega$	Section 5
Bootstrap	$1\ \mu\text{F}$ $16\ \text{V}$ X7R 0603 (TDK C1608X7R1C105K080AC)	$\geq 680\ \text{nF}$ for $0.1\ \text{V}$; droop $68\ \text{mV}$ with $2 \times 34\ \text{nC}$
Driver supply	TLV76750 LDO $12\ \text{V} \rightarrow 5\ \text{V}$, $2 \times 10\ \mu\text{F} + 100\ \text{nF}$ at VCC	gate + LDO loss $0.078\ \text{W}$ at $50\ \text{kHz}$
Local MLCCs	$12 \times 1\ \mu\text{F}$ $100\ \text{V}$ 0805	commutation loop
DC-link bank	$24 \times 10\ \mu\text{F}$ $100\ \text{V}$ 1210	$\pm 2\%$ ripple on a shared bus
Logic isolation	TI ISO7720 (shared by the inverter)	controller ground noise

4 Layout parasitics (Ansys Q3D)

The geometry of Fig. 1–2 was built in Q3D Extractor (`simulation/bb_q3d`) with the method used for the SiC busbar model: only copper is modelled, every component is a gap with source/sink terminals on its pads, and each copper region is a net (DC+, DC−, AC with 18 source and 3 sink terminals). The AC partial-inductance matrix at $100\ \text{MHz}$ is exported as an LTspice subcircuit with coupled inductors. A second design extracts the LO gate star (two branches with Kelvin returns on L2).

Commutation loop per cell, copper only (both QL switching, banks ideal)	0.312 nH
incl. MLCC ESL (local $6 \times 0.48\ \text{nH}$, bank rows $8 \times 0.5\ \text{nH}$)	0.374 nH
local MLCCs only (bank removed)	0.399 nH
with 2 oz inner layers	0.376 nH
Gate loop per FET (star branch, one FET driven / both driven)	1.70 / 2.21 nH
DC resistance, AC path (both cells, 2 oz)	0.146 m Ω
DC resistance, DC+ / DC− path from the terminals (both cells, 1 oz $\rightarrow$ 2 oz)	0.353 / 0.420 $\rightarrow$ 0.216 / 0.226 m Ω

The EPC2361 package inductance is not included. The mirrored geometry gives identical left and right values (difference $< 0.1\%$). The commutation loop is the same as the single-cell study ($0.39\ \text{nH}$ including ESL), so the second cell and the bank do not degrade it; the bank lowers it slightly. The HO star is assumed equal to the extracted LO star (it is $1\ \text{mm}$ shorter).

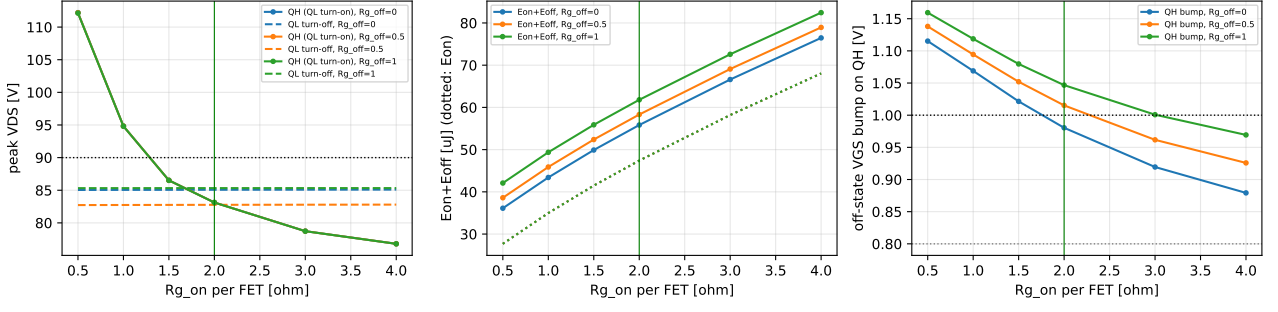
PCB copper loss

The DC+ and DC− planes carry the switching-period average current $d \cdot i$ ($65\ \text{A}_{\text{rms}}$ at the rated point) from the terminals above the bank to the cells, $15\text{--}20\ \text{mm}$ away; the AC path carries the full $100\ \text{A}_{\text{rms}}$. With the Q3D DC resistance matrix:

- 1 oz ($35\ \mu\text{m}$) inner layers: $4.67\ \text{W}$ — more than a third of the loss budget.
- 2 oz ($70\ \mu\text{m}$) inner layers (adopted): $3.30\ \text{W}$ (AC $1.46\ \text{W}$, DC+ $0.90\ \text{W}$, DC− $0.94\ \text{W}$). The loop inductance is unchanged.
- Further reductions if needed: DC busbar contacts next to each cell instead of above the bank, 3 oz outer layers for the AC path, or an AC busbar clip directly on the L1 AC islands.

5 Gate resistor selection

The LTspice model (`simulation/bb_spice`) uses the Q3D power-loop and gate-star subcircuits, four EPC2361 models, the capacitor banks with ESR/ESL, a $20\ \text{nH}$ lab busbar with $1\ \text{mF}$ bulk, and an LT8418-like driver ($0.6\ \Omega$ pull-up, $0.2\ \Omega$ pull-down, split outputs). A low-side double pulse at $160\ \text{A}$ (13% above the $141\ \text{A}$ peak) was swept over $R_{\text{g,on}} = 0.5\text{--}4\ \Omega$ and $R_{\text{g,off}} = 0\text{--}1\ \Omega$ per FET. Acceptance: peak $V_{\text{DS}} \leq 90\ \text{V}$ for both devices, off-state V_{GS} bump on the complementary FET $\leq 1.0\ \text{V}$, and $V_{\text{GS}} \geq -3\ \text{V}$; among the acceptable pairs the one with the lowest $E_{\text{on}} + E_{\text{off}}$ is chosen.

Figure 4: Gate-resistor study at 75 V / 160 A. Green line: selected $R_{g,on}$.

Selection: $R_{g,on} = 2 \Omega$, $R_{g,off} = 0 \Omega$ per FET. At 160 A: QH peak 83.1 V, QL peak 85.1 V, QH gate bump 0.98 V, minimum V_{GS} -1.29 V, $E_{on} = 47.4 \mu J$ and $E_{off} = 8.4 \mu J$ per switch position, dv/dt 12 V/ns (turn-on) and 32 V/ns (turn-off). $R_{g,on} \geq 1.5 \Omega$ is needed for the 90 V limit (1 Ω gives 95 V, 0.5 Ω 112 V), and 2 Ω for the gate-bump limit; $R_{g,off} = 0$ gives the lowest E_{off} and the smallest bump. Turn-off is load-driven at these currents ($E_{off} \approx 7-8 \mu J$ over the whole current range). The 0.98 V bump is just above the 0.8 V minimum threshold but below the 1.1 V typical; it lasts about 4 ns. Check the high-side gate on the first double-pulse board; a lower high-side gate-loop inductance is the cure if it causes shoot-through. The dv/dt values are 20–80 % averages and are affected by ringing; if a dv/dt limit is needed (motor insulation, EMI; EPC9186 limits to 10 V/ns), $R_{g,on}$ is the knob, at the cost of E_{on} .

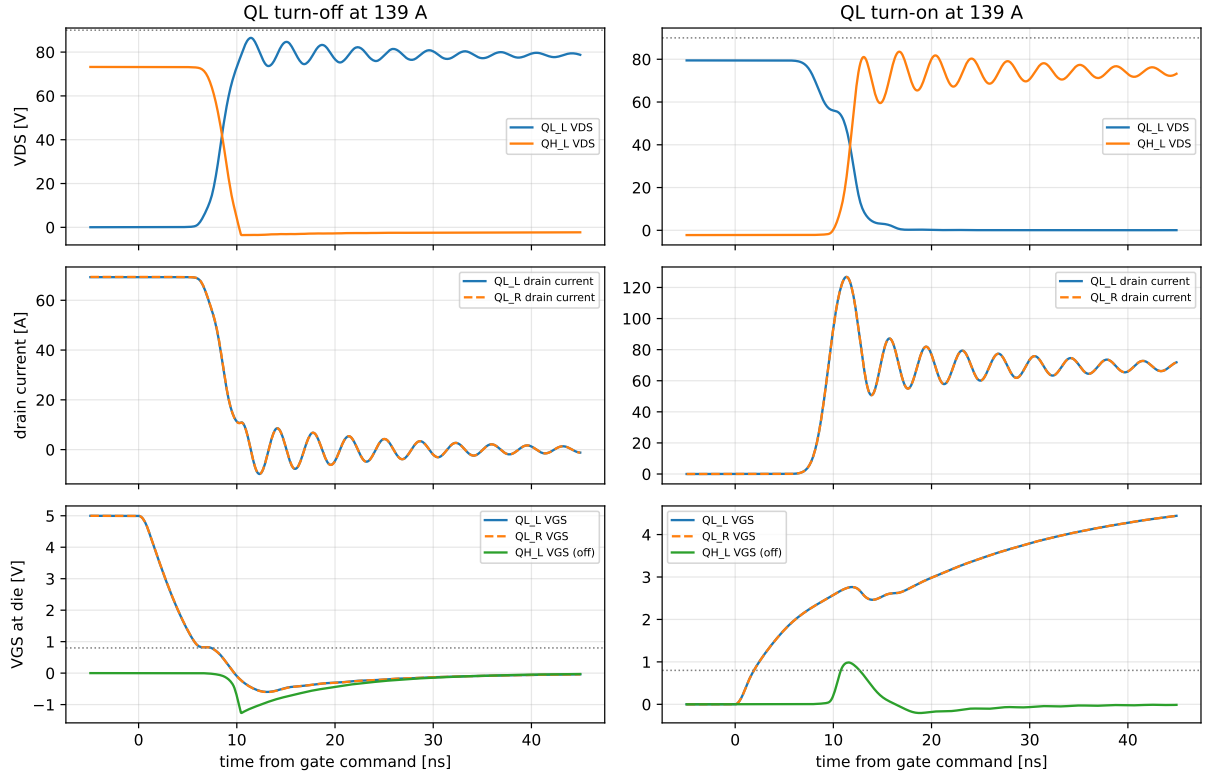
Building block, 75 V, $R_{g,on}$ 2 ohm / $R_{g,off}$ 0 ohm, Q3D parasitics (left and right FETs overlap)

Figure 5: Switching at the rated peak current with the selected resistors. The left and right FETs (solid and dashed) overlap: the symmetric layout shares current equally at both edges.

6 Losses and efficiency

The switching energy versus current was simulated with the selected resistors and fitted with second-order polynomials; the inverter switching loss is $P_{sw} = f_{sw} \frac{1}{\pi} \int_0^\pi [E_{on} + E_{off}] (I_{pk} \sin \theta) d\theta$ (average 38.1 μJ per period).

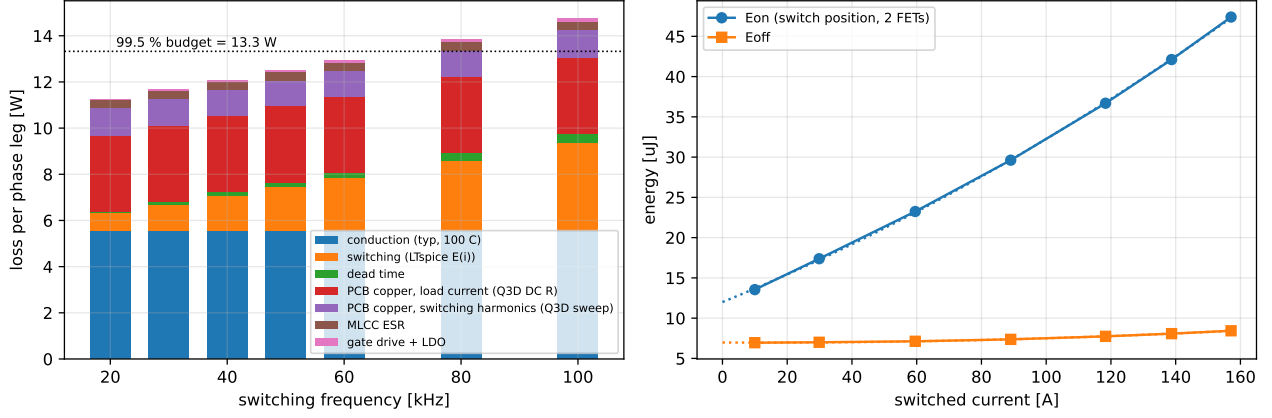


Figure 6: Left: loss per phase leg versus switching frequency at the rated point. Right: switching energy per switch position (two FETs) versus current, LTspice with Q3D parasitics.

Loss component at 50 kHz, rated point	W
Conduction, $R_{DS(on)}$ typ at 100 °C ($I^2 R_{DS}/2$)	5.55
Switching (LTspice E(i), selected R_g)	1.91
Dead time (2×10 ns per period, reverse conduction)	0.20
PCB copper, load current (Q3D DC resistance)	3.30
PCB copper, switching harmonics (Q3D frequency sweep)	1.07
MLCC ESR (network solve)	0.39
Gate drive + LDO	0.078
Total	12.49
Efficiency (budget 13.3 W)	99.53 %
With $R_{DS(on)}$ max at 100 °C	14.34 W, 99.46 %

The 99.5 % target is met up to 60 kHz with typical devices. With maximum $R_{DS(on)}$ it is missed at 50 kHz (14.34 W, 99.46 %). The switching-harmonic copper loss comes from a Q3D frequency sweep (1 kHz–100 MHz): the ripple current between the MLCC bank and the cells sees about $1.5 \times$ the DC resistance at 50 kHz; a nodal solve of the board network at every harmonic of the SVM switching currents gives 1.07 W at the rated point and up to 2.5 W at low modulation index (`scripts/bb_copper_hf.py`). E_{on} does not go to zero at zero current: charging the high-side output capacitance through the low-side channel costs about $Q_{oss} V \approx 2 \times 113 \text{ nC} \times 75 \text{ V}$. Conduction dominates; at 25 °C it would be 3.75 W, so cooling directly reduces loss. Per device the loss is about 1.91 W at 50 kHz; with 0.2 K/W junction-to-case, an insulating TIM of about 1.9 K/W per device and a 60 °C heat sink, $T_j \approx 64$ °C. The conduction loss above assumes 100 °C, so this is conservative.

7 Limitations and next steps

- MLCC capacitance under 75 V DC bias is assumed (30 % / 45 %); confirm with Murata SimSurfing and TDK data before freezing the bank.
- The Q3D model is a pre-layout geometry with simplified two-bar FET pads; repeat the extraction on the routed KiCad board. Package inductance and frequency-dependent resistance are not in the SPICE model, so ringing is under-damped.
- Losses use typical device parameters and a 10 ns dead time; switching energy comes from a model, not a measurement. Validate with the double-pulse board.
- Thermal design (TIM, heat sink, coolant) is only estimated; the per-device budget above sets its requirement.

Files. Sizing `scripts/bb_sizing.py`; geometry simulation `bb_q3d/bb_geometry.py`; Q3D simulation `bb_q3d/bb_q3d.py`; LTspice simulation `bb_spice/`; results and this report's numbers `scripts/bb_losses.py`; layout page docs/building-bloc